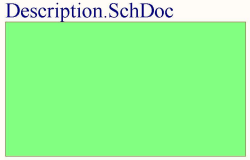
Description.SchDoc



PowerDistribution.SchDoc



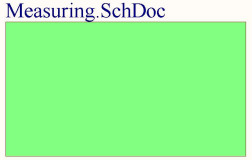
MicroController.SchDoc



GateDriverandMosfets.SchDoc



Measuring.SchDoc



Control.SchDoc



Title: ***Main***		*Mergen Elektronik*	MERGEN
Size: A4 Number: 20240101	Revision: v0.0.1	Türkiye	
Date: 26.10.2024 Time: 00:18:16	Sheet 1 of 7	* *	Drawn By : ~~Ekrem T. Fırat~~
Project Mergen SMPS Board.PrjPcb		*	

Mergen Switch Mode Power Supply Board

Attention: This project is open source and still under development. The project is always dangerous because it involves grid electricity's transformation and control. The engineer is not responsible for any adverse situation that may occur.
Contact: ig:mergenelk

Project Purpose

This project aims to use 220VAC in 0-30V direct current with transformer and converter circuits.

Power Distribution

The scheme aims to filter 36VAC from an externally connected transformer and convert it into Direct Current to provide the necessary voltage to the components and the system.

MicroController

In this scheme, it is aimed to produce PWM, make the necessary measurements, reflect it on the screen and provide the output.

Gate Driver and MOSFETS

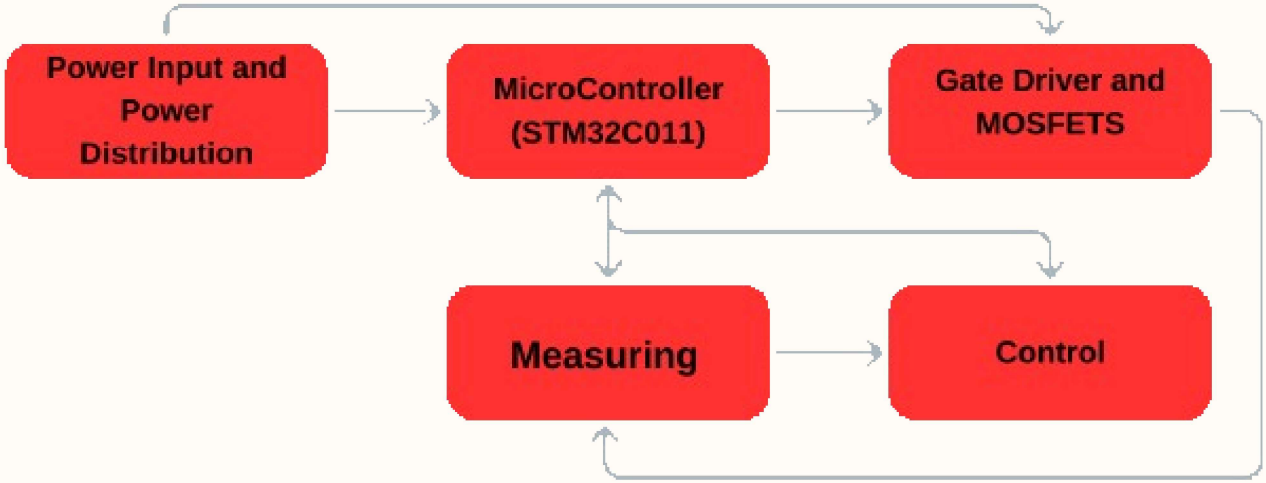
This scheme is intended to control the MOSFETs and filter the power output.

Measuring

This scheme is intended to measure the output current and voltage and ensure that the current limit is not exceeded.

Measuring

This diagram is intended to set the desired output voltage and current limit.



Port Colors

- VAC Input
- Relay
- Control
- LCD
- Gates I/O
- Measuring

Some port colors are left as default.

GND, VCC, 7.5V, 3.3V lines are specially used as thick and colored.

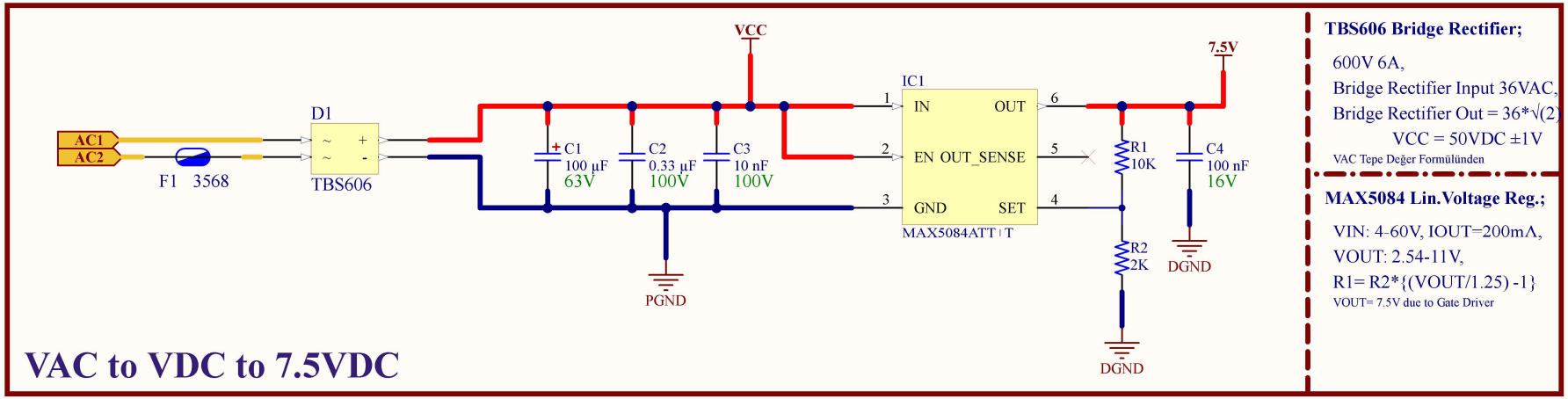
Resources

1) Basic Calculation of a Buck Converter's Power Stage, August 2015, SLVA447B - Texas Instruments
2) Understanding Buck Power Stages in Switchmodes Power Supplies, March 1999, SLVA057 - Texas Intruments
3) STM32C011 Datasheet, January 2024, STM32C011x4/x6 - STMicroelectronics
4) UCC27282 Datasheet, May 2022, UCC27282 - Texas Instruments
5) Best of Power Tips, SSZTCY0 - Texas Instruments
6) and component datasheets :)

Thank you to everyone who provided their information and opinions.

Title: *Description*		*Mergen Elektronik*
Size: A4	Number: 20240101	Revision: v0.0.1
Date: 26.10.2024	Time: 00:18:16	Sheet 2 of 7
Project Mergen SMPS Board.PrjPcb		Türkiye
		Drawn By : Ekrem T. Fırat

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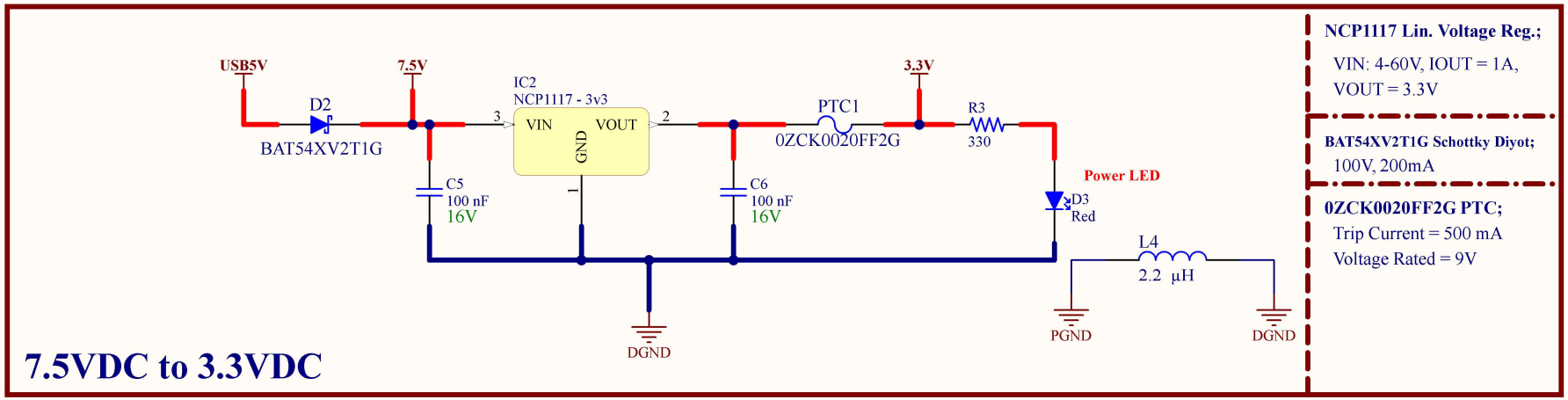


TBS606 Bridge Rectifier;

600V 6A,
Bridge Rectifier Input 36VAC,
Bridge Rectifier Out = 36*√(2)
VCC = 50VDC ±1V
VAC Tepe Değer Formülünden

MAX5084 Lin.Voltage Reg.;

VIN: 4-60V, IOUT=200mA,
VOUT: 2.54-11V,
R1= R2*{(VOUT/1.25) -1}
VOUT= 7.5V due to Gate Driver



NCP1117 Lin. Voltage Reg.;

VIN: 4-60V, IOUT = 1A,
VOUT = 3.3V

BAT54XV2T1G Schottky Diyot;

100V, 200mA

0ZCK0020FF2G PTC;

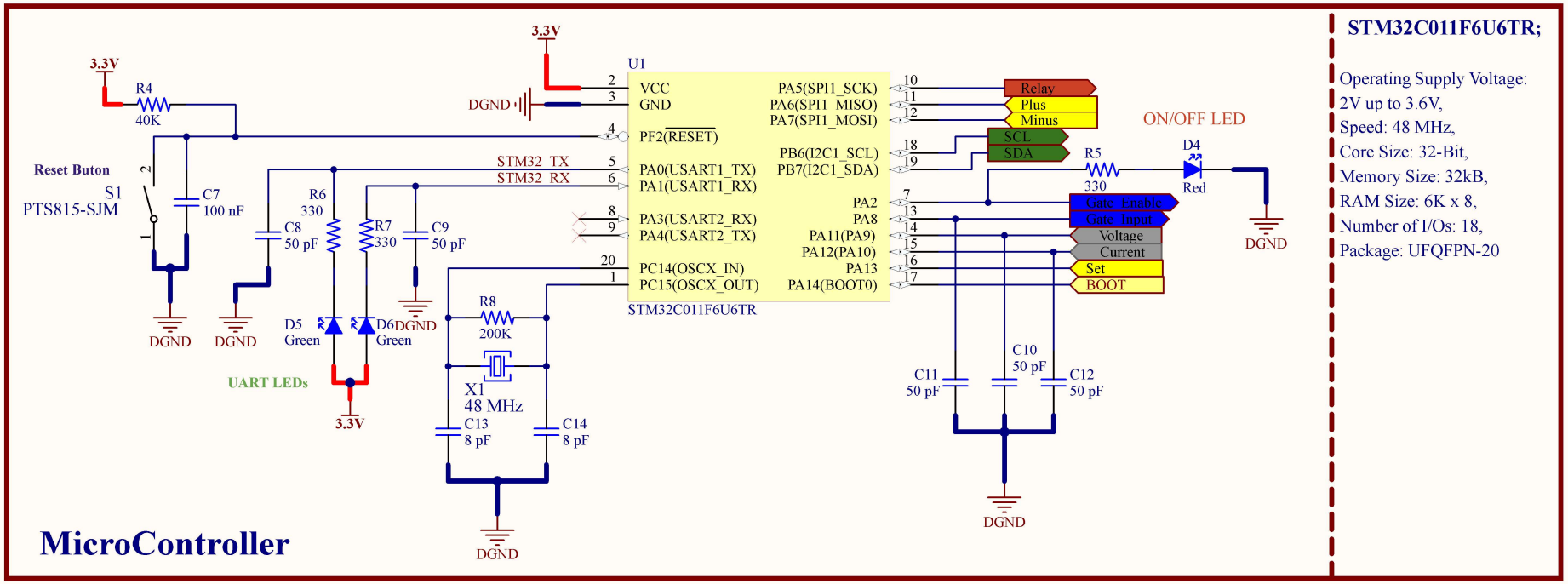
Trip Current = 500 mA
Voltage Rated = 9V

Title: ***Power Distribution***		*Mergen Elektronik*
Size: A4 Number: 20240101	Revision: v0.0.1	Türkiye
Date: 26.10.2024 Time: 00:18:16	Sheet 3 of 7	* *
Project Mergen SMPS Board.PrjPcb		*

MERGEN

Drawn By : ~~Ekrem T. Fırat~~

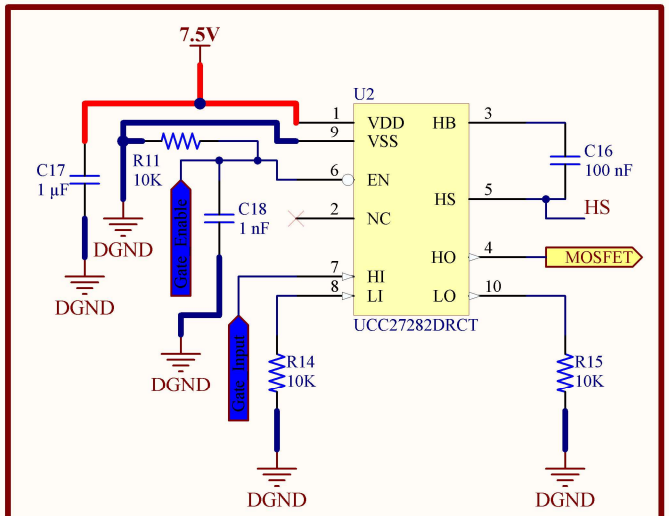
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STM32C011F6U6TR;

Operating Supply Voltage:
2V up to 3.6V,
Speed: 48 MHz,
Core Size: 32-Bit,
Memory Size: 32kB,
RAM Size: 6K x 8,
Number of I/Os: 18,
Package: UFQFPN-20

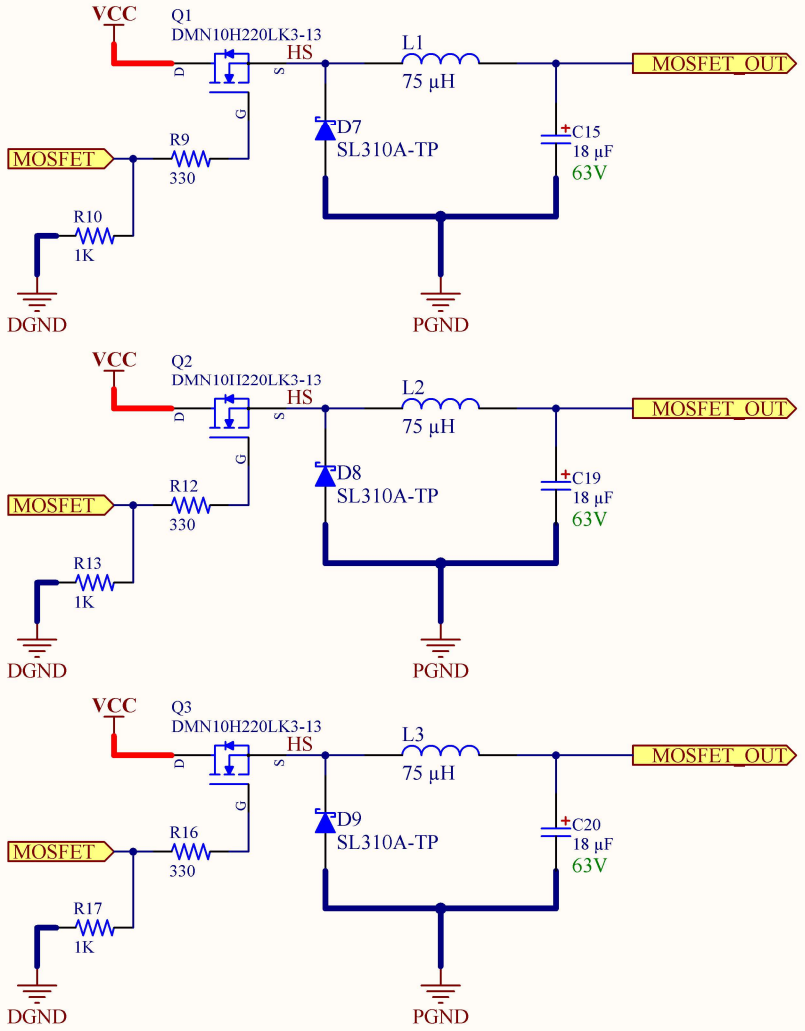
Title: *MicroController*		*Mergen Elektronik*
Size: A4 Number: 20240101	Revision: v0.0.1	Türkiye
Date: 26.10.2024 Time: 00:18:16	Sheet 4 of 7	* * *
Project Mergen SMPS Board.PrjPcb		Drawn By : Ekrem T. Fırat



Gate Driver Circuit

UCC27282DRCT Half Bridge Gate Driver;

Operating Supply Voltage: 5.5V up to 16V,
Boot Voltage: 120VDC, Output Peak Current: 3A,
Gate Type: N Channel,
Integreted Boostrap Diode



MOSFETS

Switching MOSFETS

Productivity: %80(estimated) n
Max. Input Voltage: 51
Min. Input Voltage: 50V Vin
Output Voltage: 0-30V Vout
Max. Output Current: 5A Iout
Frequency: 100 kHz F
Max. Duty Cycle: D
Vout/(Vin * n) = 30/(50*0.6) = %75

Inductor Ripple Current: ΔIL
0.3*Iout(max) = 0.3*5 = 1.5A
Inductor Inductance Value: L
Vout*(Vin - Vout)/(ΔIL*F*Vin)
30*(20)/(1.5*100000*50) = 80µH
Capacitor Value:
ΔIL/(8*F*ΔVout) ΔVout = ΔIL*ESR
1.5/(8*100000*0.09) = 20 µF
Diode Current:
Iout*(1-D) = 5*(1-0.75) = 1.25A

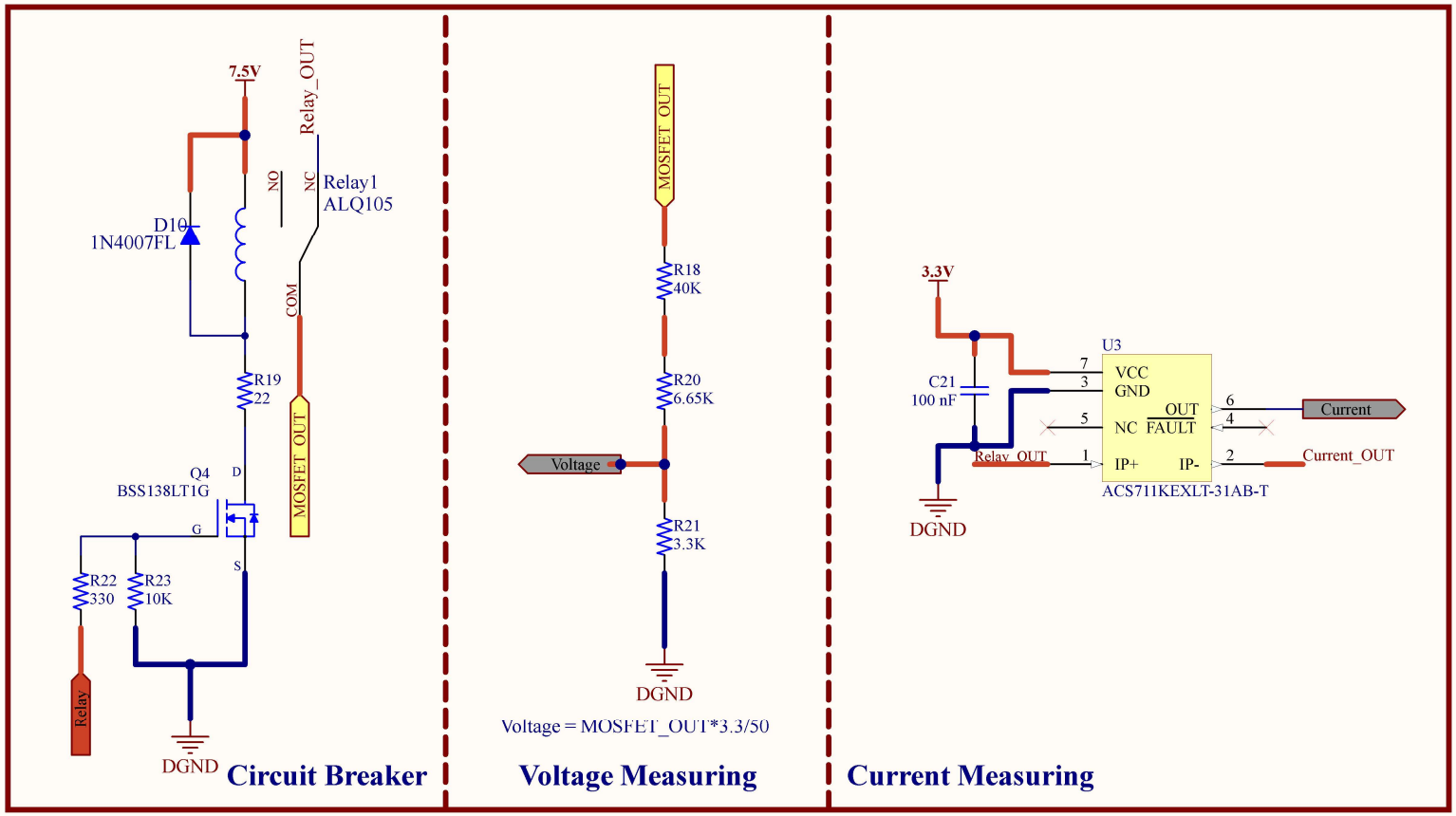
MOSFET Max. Output Current
Id - ΔIL/2 = 7.5 - 0.75 = 6.75A
MOSFET Switching Current
Iout + ΔIL/2 = 5 + 0.75 = 5.75A

Note: The components closest to the calculations are preferred.

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Contact: ig:mergenelk

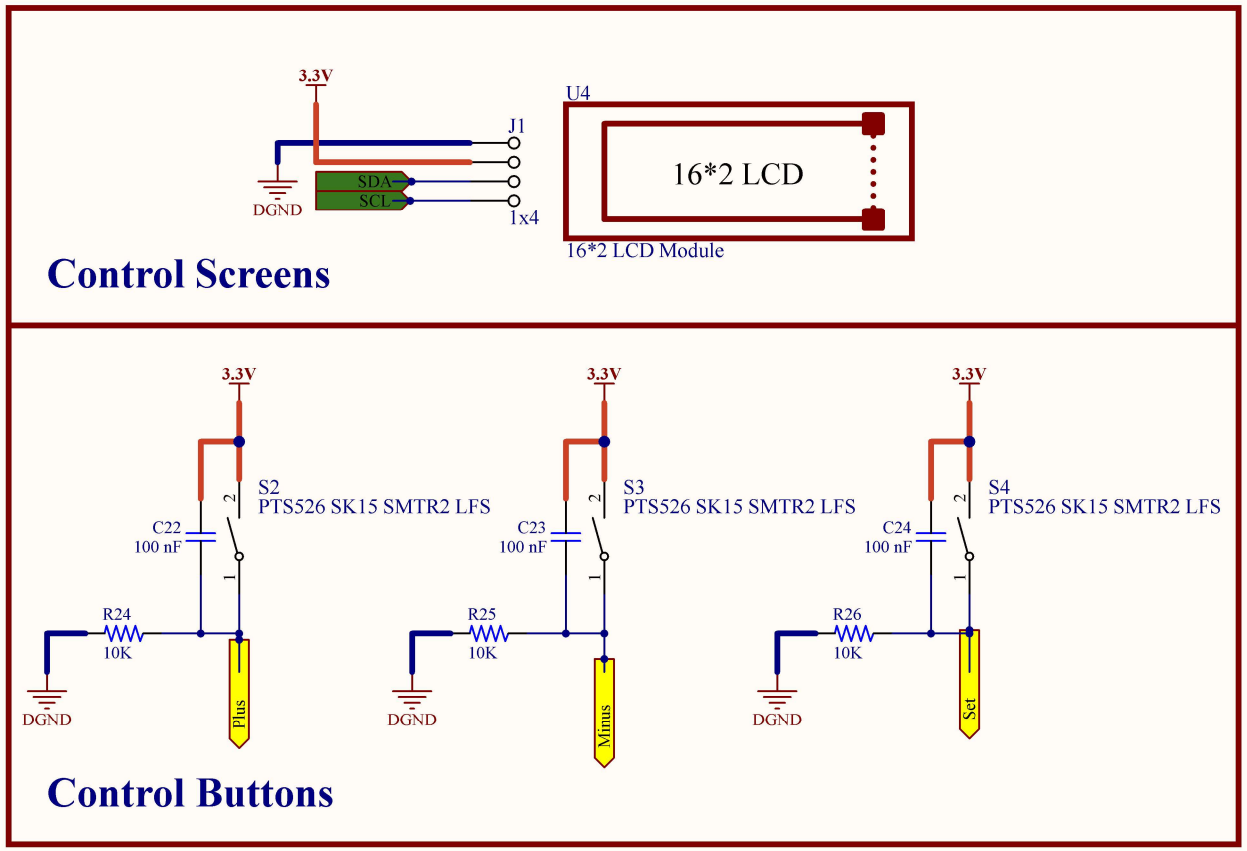
Title: *Gate Driver and MOSFETs*		*Mergen Elektronik*	MERGEN
Size: A4	Number: 20240101	Revision: v0.0.1	Türkiye
Date: 26.10.2024	Time: 00:18:16	Sheet 5 of 7	Drawn By : Ekrem T. Fırat
Project Mergen SMPS Board.PrjPcb			

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Contact: ig:mergenelk



Title: ***Measuring***		*Mergen Elektronik*	MERGEN
Size: A4	Number: 20240101	Revision: v0.0.1	Türkiye
Date: 26.10.2024	Time: 00:18:16	Sheet 6 of 7	Drawn By : Ekrem T. Fırat
Project Mergen SMPS Board.PrjPcb			

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Contact: ig:mergenelk



Title: *Control*		*Mergen Elektronik*	MERGEN
Size: A4	Number: 20240101	Revision: v0.0.1	Türkiye
Date: 26.10.2024	Time: 00:18:17	Sheet 7 of 7	Drawn By : Ekrem T. Fırat
Project Mergen SMPS Board.PrjPcb			